

SafeDrive Insurance — Motor Insurance Claims Risk Dashboard

Capstone Analytics Project · Stage 4 Dashboard Report

Page 1 · Portfolio Overview

Total policies

58,592

Total claims

3,748

Claim rate

6.4%

High-risk customers

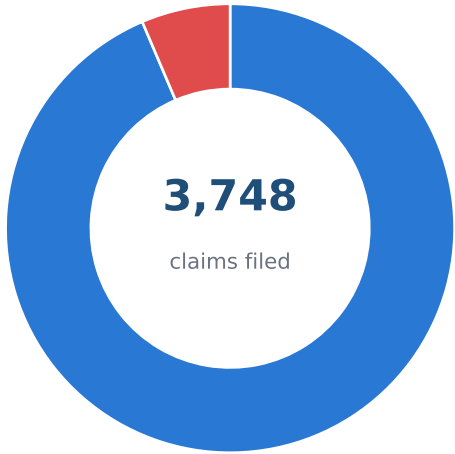
3,054

of 11,719 scored test-set policies

Average NCAP rating

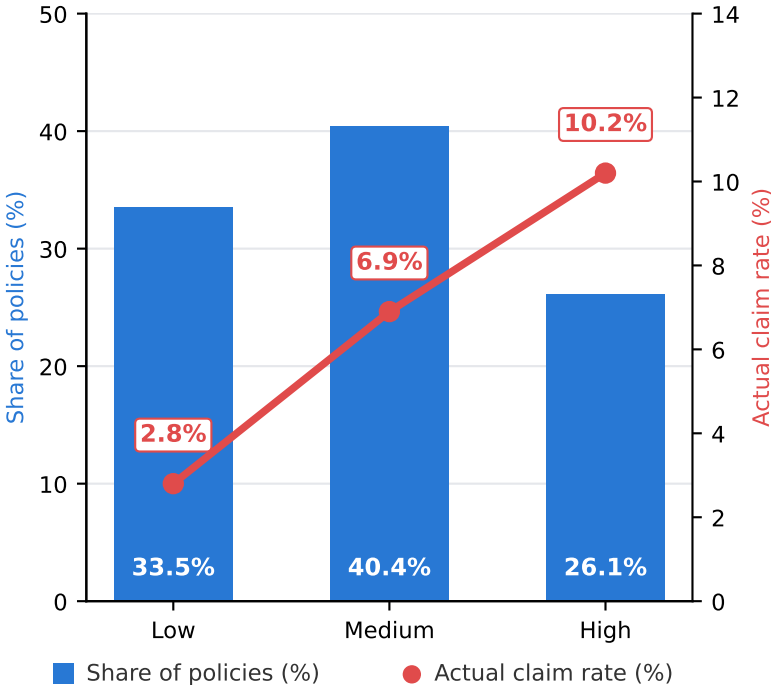
1.76 / 5

Claim distribution



■ No claim 93.6% ■ Claim filed 6.4%

Driver risk segmentation (test set, n=11,719)



Model performance (test set)

Metric	Logistic Reg.	Decision Tree
ROC-AUC	0.596	0.651
Recall	0.560	0.683
Precision	0.081	0.094
F1 score	0.142	0.164

Logistic Reg. = baseline model.
Decision Tree = Optuna-tuned final model, better on recall and ROC-AUC.

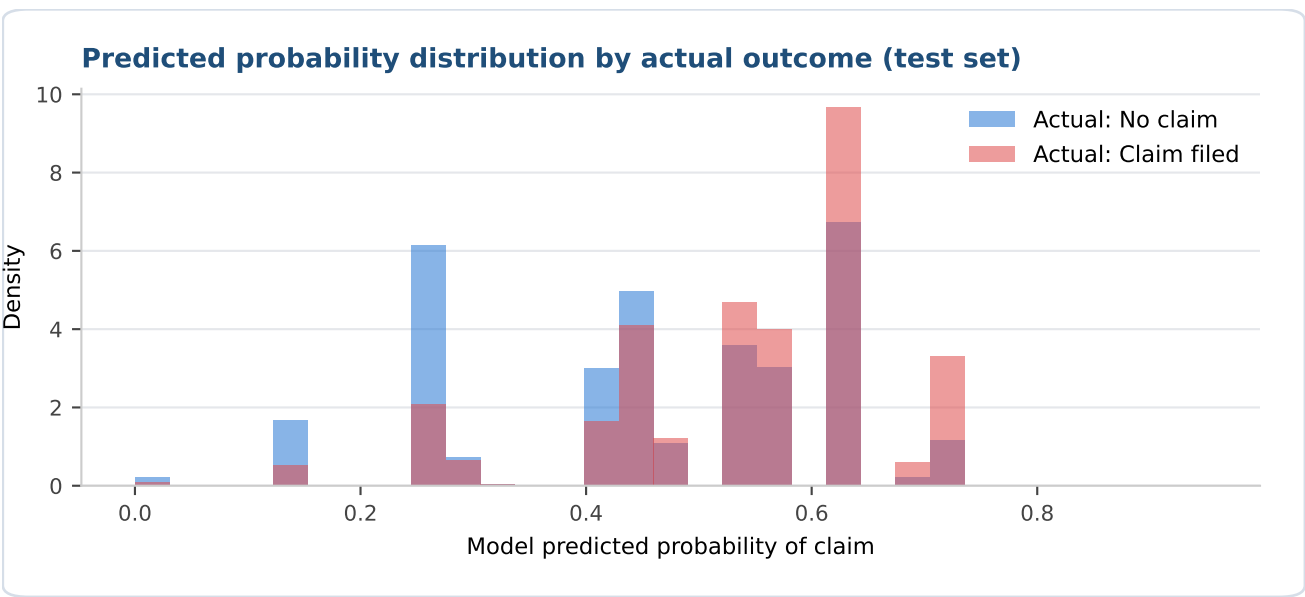
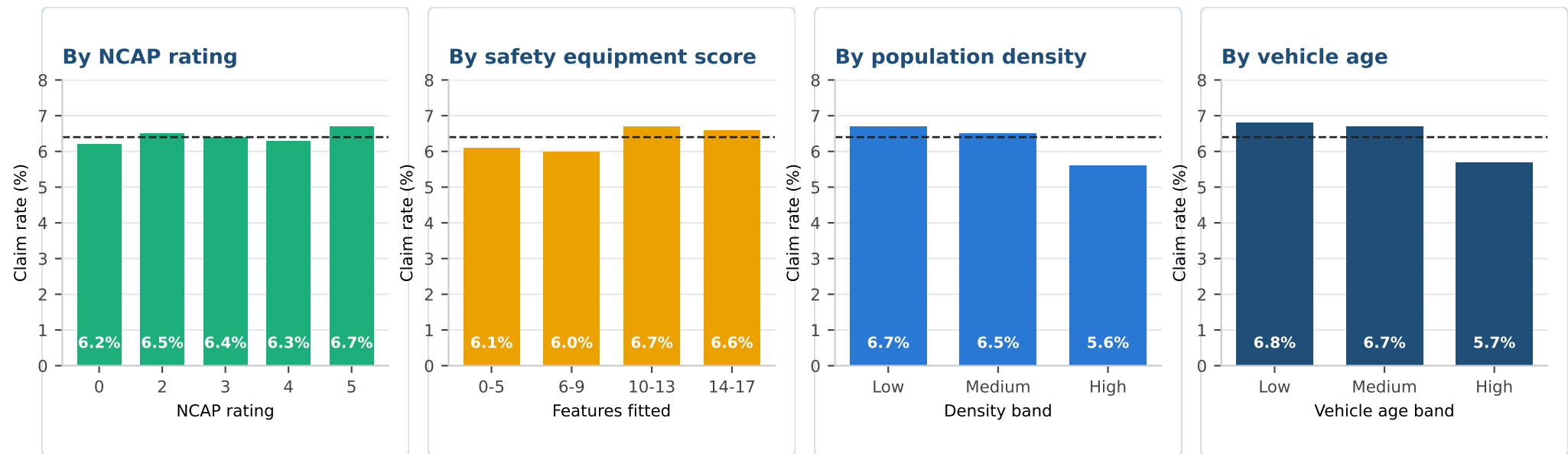
Risk tiers and probability scores come from the final tuned Decision Tree model (Stage 3), applied to the 20% held-out test set (11,719 policies).
The descriptive views on Page 2 use the full 58,592-policy book.

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Page 2 · Risk Drivers & Model Output

Claim rate by key risk drivers (full 58,592-policy book)



Key insights

- No single feature separates claimants cleanly: claim rates only range from about 5.6% to 6.9% across NCAP rating, safety score, density and vehicle age.
- Risk tiers do separate claims: High-risk drivers claim at 10.2% vs 2.8% for Low-risk (about 3.6x).
- Test ROC-AUC is 0.65, a modest but real lift. This is expected given rare claims (6.4%) and weak individual predictors.
- The risk-tier view (Page 1) is the most decision-useful output of the model.

Predicted probabilities and tiers are from the tuned Decision Tree (Stage 3) on the 20% held-out test set. Dashed line = overall claim rate (6.4%).